

3.2.2 Authentication API

An account is used to call the login API to obtain a token for identity authentication when accessing other data query APIs and control APIs.

3.2.2.1 Login API

API Description

- Before obtaining data, the login API must be called to obtain the XSRF-TOKEN. The validity period of XSRF-TOKEN is 30 minutes.
- If the XSRF-TOKEN does not expire, it can be reused. If the XSRF-TOKEN has expired, the login API needs to be called again to obtain a new XSRF-TOKEN.
- After this API is called to log in to the system, XSRF-TOKEN is returned in the response header.
- A new token is generated each time this API is called, and the previously obtained token becomes invalid.

Request URL

https://Domain name of the management system/thirdData/login

Request Mode

HTTP method: POST

Access Restrictions

If an API account enters incorrect passwords for five consecutive times within 10 minutes, the API account will be locked out for 30 minutes.

Maximum number of API calls for each API account: five times every 10 minutes.

If the access frequency exceeds the limit, the API returns error code 407.

Request Parameters

Parameter	Description	Data Type	Mandatory/Optional
userName	API account name	String	Mandatory
systemCode	Password	String	Mandatory

Response Packet

Parameter		Description	Data Type	Remarks
success		Request success or failure flag. true: The request succeeded. false: The request failed.	Boolean	Request success or failure flag
failCode		Error code. Value 0 indicates that the status is normal. For other error codes, see 6.1 Error Code List .	Integer	-
params	Parameters	-	-	-
	currentTime	Current system timestamp, in milliseconds.	Long	Example : 1501862400000
message		Optional message.	String	-
data		Returned data.	Object	-

Examples

Request example:

```
{
  "userName":"*****",
  "systemCode":"*****"
}
```

Response examples:

Example 1: successful login

```
{
  "data": null,
  "success": true,
  "failCode": 0,
  "params": {
  },
  "message": "Northbound API Reference: https://info.support.huawei.com/*****/#/u?i=UsWJrsyw&l=en_US"
}
```

Example 2: failed login

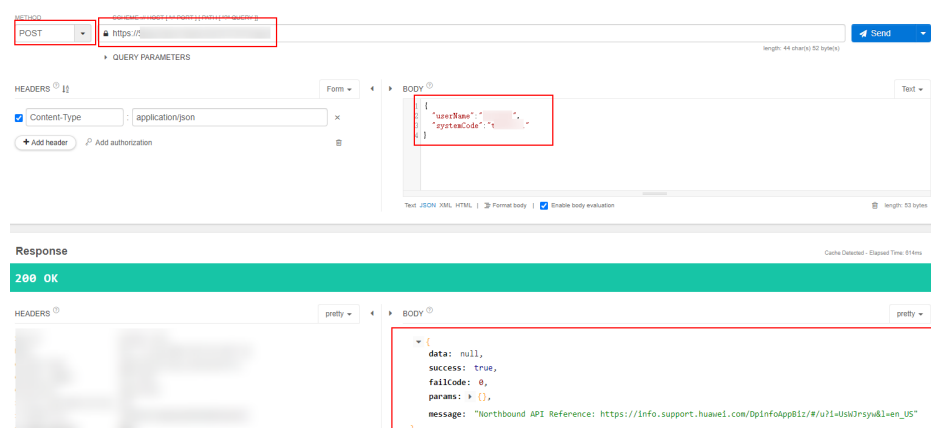
```
{
  "data": null,
  "success": false,
  "failCode": 20400,
  "params": {
    "currentTime": 1725072482769
  },
}
```

```
} "message": "user.login.user_or_value_invalid"
```

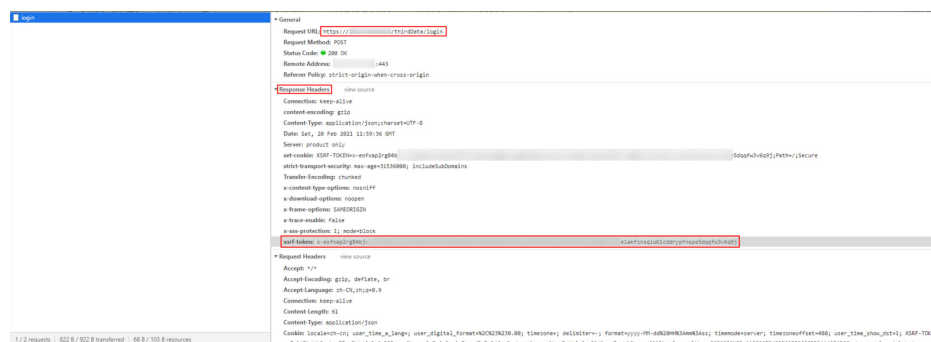
NOTICE

The header of the login success response contains the XSRF-TOKEN that must be retained. In subsequent data API requests, this parameter and its value must be added to the request header and sent to the management system.

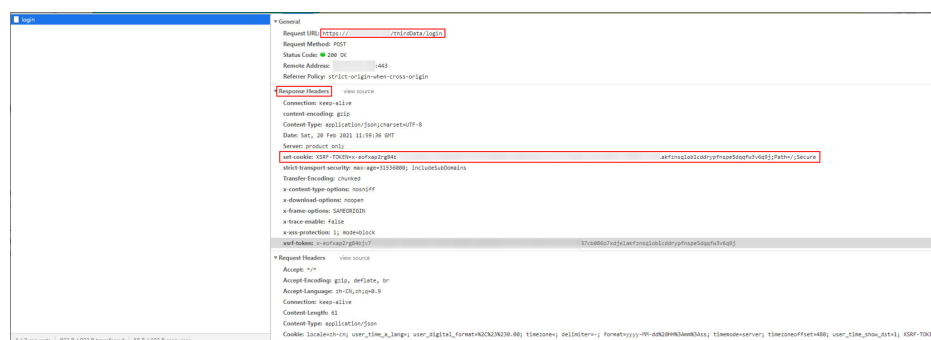
Login example:



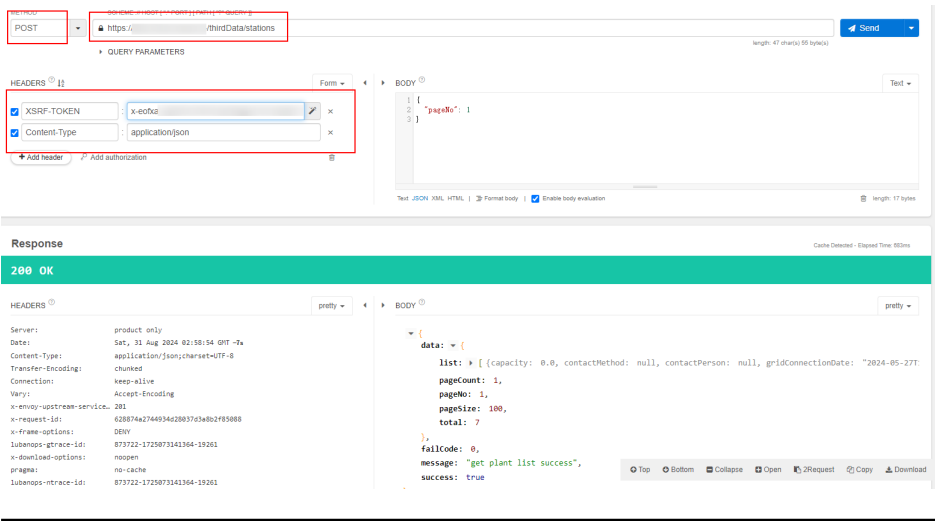
The following are examples of the XSRF-TOKEN returned after a successful login. You can obtain the XSRF-TOKEN using either of the following methods. The first one is recommended.



The following is an example compatible with earlier versions.



The following figure shows an example of the XSRF-TOKEN carried in the request header of the data API.



3.2.2.2 Logout API

API Description

If you want the XSRF-TOKEN to expire immediately, you can call this API.

Request URL

https://Domain name of the management system/thirdData/logout

Request Mode

HTTP method: POST

Access Restrictions

Maximum number of API calls for each API account: five times every 10 minutes.

You are advised to call this API only when necessary.

If the access frequency exceeds the limit, the API returns error code 407.

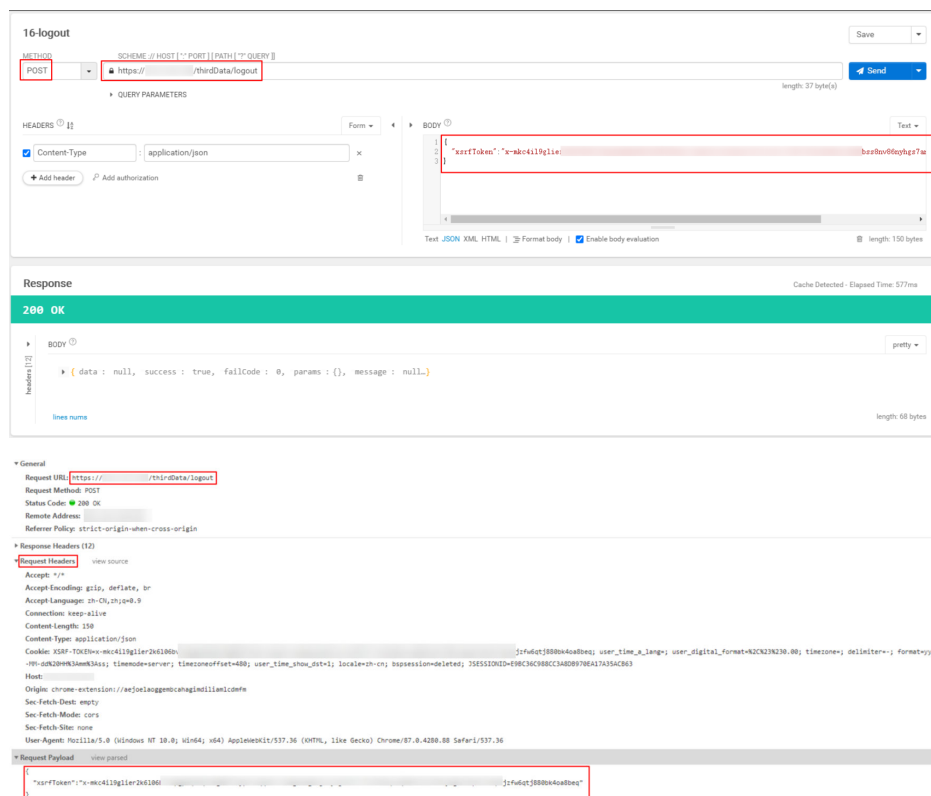
Request Parameters

Parameter	Description	Data Type	Mandatory/Optional
xsrftoken	XSRF-TOKEN returned in the response header after a successful login through the login API.	String	Mandatory

```
},  
"message":null  
}
```

NOTE

Logout example:



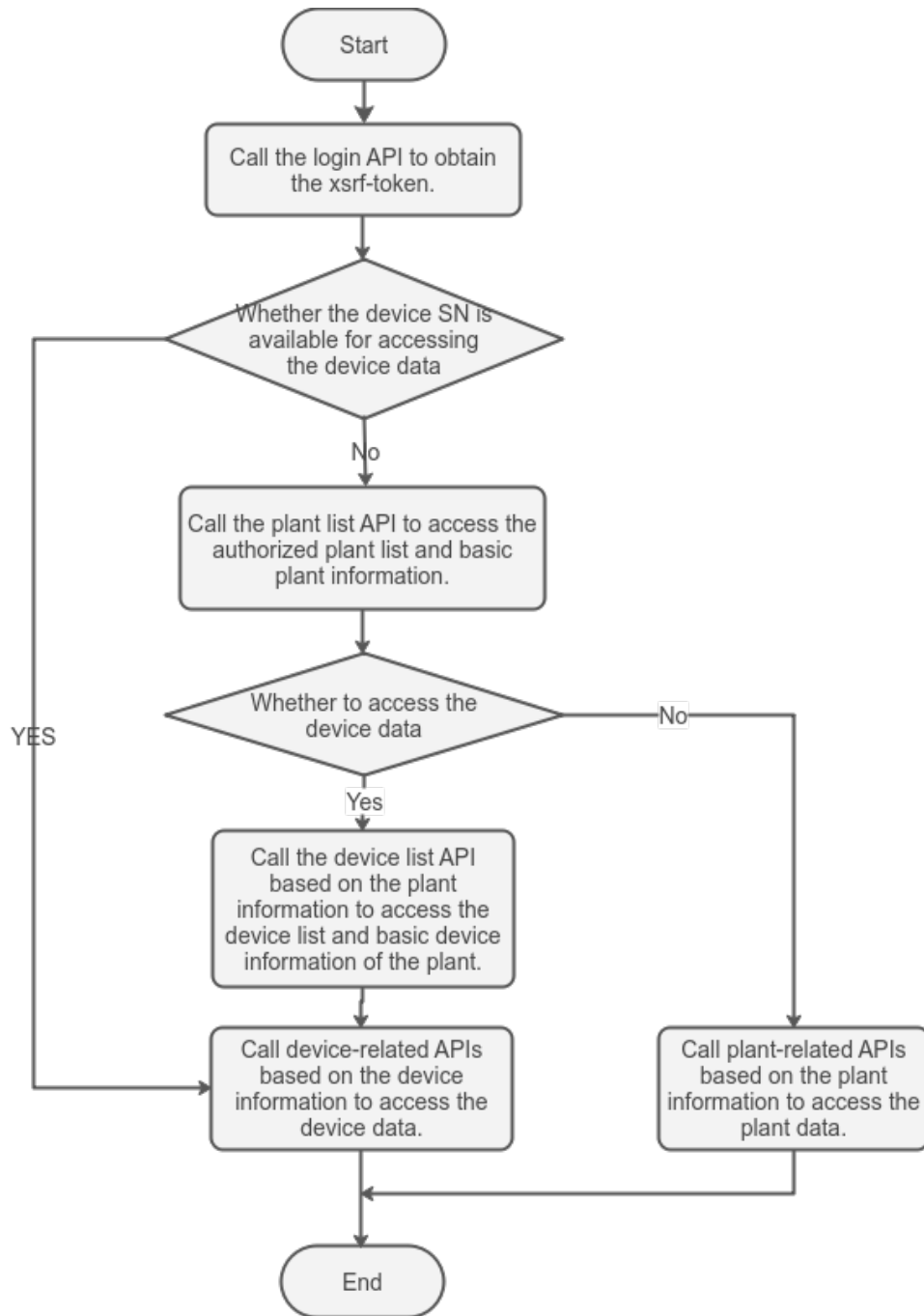
3.2.3 Calling an API

API Access

https://Domain name of the management system/specific API name+access request parameters

Access Process

Log in to the API on the third-party system and obtain the xsrf-token used for identity authentication. The xsrf-token must be carried when specific data is accessed. The third-party app calls APIs to access the specific data as required in the following process.



Access Example

- Step 1** Create an API account on the management system and bind the account to a PV plant. For details, see [3.2.1 Obtaining an Account](#).
- Step 2** Obtain the login credential xsrf-token by referring to [3.2.2.1 Login API](#). The xsrf-token can be reused if it is valid. For details about the validity period of the xsrf-token, see [API Description](#).
- Step 3** Content-Type: application/json
Accept: application/json, */*

Enter the login credential `xsrf-token` in the request header to access the corresponding API to obtain data.

[illegible]

----End

4 Flow Control Description

The system provides the API flow control mechanism to prevent system performance deterioration caused by improper API invoking. If the maximum flow volume is exceeded, this API cannot be called and the error code 407 is returned.

NOTICE

The flow control mechanism may be modified with the system evolution in the future without notice. Obtain the latest API document to view the API flow control mechanism.

When the API flow is limited, the system displays the reason for the flow control through the error code (failCode). Wait until the API returns a normal response and call the API again.

- Error code 407: The number of API calls of a single user exceeds the upper limit. When this error code is returned by an API, lower your frequency of calls to this API according to the following table until the frequency drops to the allowed range.
- Error code 429: When a large number of users initiate calls to the same API, the total number of calls to the API exceeds the upper limit at the system level. If this error message is received, wait for 1 minute and try again. If the error persists, wait for a longer time and try again. You can also minimize the number of calls.

Northbound open APIs support the following access modes. For details about flow control in OAuth Connect mode, see [4.1 Flow Control Policy in OAuth Connect Mode](#). For details about flow control using an API account, see [4.2 Flow Control Using the API Account](#).

4.1 Flow Control Policy in OAuth Connect Mode

The API flow control is accumulated based on the API category. Call APIs properly based on the flow control description.

API Category	Flow Control Description
Basic APIs	1000 times/day/owner. The number of flow control times is counted independently for each owner. Example: • If an owner authorizes a third-party app, the third-party app can access the owner's resources through the API for up to 1000 times a day. • If 100 owners authorize a third-party app, the third-party app can access the resources of each owner through the API for up to 1000 times a day.
Control APIs	100 times/day/owner. The number of flow control times is counted independently for each owner. Example: • If an owner authorizes a third-party app, the third-party app can access the owner's resources through the API for up to 100 times a day. • If 100 owners authorize a third-party app, the third-party app can access the resources of each owner through the API for up to 100 times a day.

4.2 Flow Control Using the API Account

The following table describes the flow control of unrestricted northbound APIs. Call APIs properly based on the flow control description.

API Category	API Label	API Name	Flow Control Description
Basic APIs	Basic data	Plant List API	Maximum number of API calls for each API account per day = Roundup (Number of plants/100) x 10 + 24 Example: • If an API account manages 20 plants, the maximum number of API calls per day = Roundup (20/100) x 10 + 24 = 1 x 10 + 24 = 34. • If an API account manages 120 plants, the maximum number of API calls per day = Roundup (120/100) x 10 + 24 = 2 x 10 + 24 = 44.

API Category	API Label	API Name	Flow Control Description
		Device List API	Flow control is performed based on the number of plants managed by API accounts. Maximum number of API calls for each API account per day = Roundup (Number of plants/100) + 24 Example: • If an API account manages 20 plants, the maximum number of API calls per day = Roundup (20/100) + 24 = 1 + 24 = 25. • If an API account manages 120 plants, the maximum number of API calls per day = Roundup (120/100) + 24 = 2 + 24 = 26.
	Monitoring data	API for Real-time Plant Data	Flow control is performed based on the number of plants managed by API accounts. Maximum number of API calls for each API account every 5 minutes = Roundup (Number of plants/100) Example: • If an API account manages 20 plants, the maximum number of API calls every 5 minutes = Roundup (20/100) = 1 • If an API account manages 120 plants, the maximum number of API calls for each API account every five minutes = Roundup (120/100) = 2

API Category	API Label	API Name	Flow Control Description
		Historical Device Data API	Flow control is performed based on the number of devices of each type managed by an API account. Maximum number of API calls for each API account per day = $\sum \text{Roundup}(\text{Number of devices of each type}/10) + 24$. Example: • If an API account manages 20 inverters and 20 meters, the maximum number of API calls per day is calculated as follows: Inverters: $\text{Roundup}(20/10) = 2$ Meters: $\text{Roundup}(20/10) = 2$ Total: Maximum number of API calls for inverters + Maximum number of API calls for meters = $2 + 2 + 24 = 28$ • If an API account manages 120 inverters and 120 meters, the maximum number of API calls per day is calculated as follows: Inverters: $\text{Roundup}(120/10) = 12$ Meters: $\text{Roundup}(120/10) = 12$ Total: Maximum number of API calls for inverters + Maximum number of API calls for meters = $12 + 12 + 24 = 48$

5 API Reference

5.1 Basic APIs

Allow you to obtain the real-time monitoring, report, and alarm data of plants and devices for system monitoring, O&M, and data analysis.

5.1.1 Basic

5.1.1.1 Plant List API

API Description

This API is used to query the plant list. When the pagination parameters (a maximum of 100 records can be displayed on each page) and grid connection time are transferred (when only the grid connection start time is transferred, the grid connection end time is the current time by default. When only the grid connection end time is transferred, the default grid connection start time is the start point of the timestamp, and the plant list is queried in pages based on the grid connection time. When only the pagination parameter is transferred, the plant list is queried in pages.)

CAUTION

If no data is queried or only some data is returned when the plant list API is used, rectify the fault by referring to [8.3 Why Is No Data or Only Part of Data Found When I Call a Northbound API for Data Query?](#)

If the returned plant ID is inconsistent with that displayed on the SmartPVMS, rectify the fault by referring to [8.7 Why Is the Plant ID Returned by the Plant List Inconsistent with That Displayed on the SmartPVMS?](#)

If the plant IDs returned vary depending on API accounts, rectify the fault by referring to [8.8 Why Do the Plant IDs Returned for the Same Plant Vary Depending on API Accounts?](#)

Request URL

<https://Domain name of the management system/thirdData/stations>

Request Mode

HTTP method: POST

Request Parameters

Parameter	Description	Data Type	Mandatory/Optional
pageNo	Page No. of the results.	Integer	Mandatory
gridConnectedStartTime	Grid connection start timestamp (in milliseconds), for example, 1501862400000.	Long	Optional
gridConnectedEndTime	Grid connection end timestamp (in milliseconds), for example, 1501862400000.	Long	Optional

Response Packet

Parameter	Description	Data Type	Remarks
success	Request success or failure flag. true: The request succeeded. false: The request failed.	Boolean	Request success or failure flag
failCode	Error code. Value 0 indicates that the status is normal. For other error codes, see 6.1 Error Code List .	Integer	-
message	Optional response message.	-	-
data	Returned data, which contains the following information:	Map	-
> total	Total number of results.	Long	-
> pageCount	Total number of pages.	Long	-

Parameter	Description	Data Type	Remarks
> pageNo	Page No. of the results.	Integer	-
> pageSize	Number of query results displayed on each page.	Integer	-
> list	Plant information list. The plant information is as follows:	List	Plant information
>> plantCode	Plant ID, which uniquely identifies a plant.	String	-
>> plantName	Plant name.	String	-
>> plantAddress	Detailed address of the plant.	String	-
>> longitude	Plant longitude.	Double	-
>> latitude	Plant latitude.	Double	-
>> capacity	Total string capacity.	Double	kWp
>> contactPerson	Plant contact.	String	-
>> contactMethod	Plant contact information, such as the mobile number or email address.	String	-
>> gridConnectionDate	Grid connection time of the plant, including the time zone.	String	2020-02-06T00:00:00+08:00

Examples

Request example:

```
{
  "pageNo": 1,
  "gridConnectedStartTime":1664718569000,
  "gridConnectedEndTime":1667396969000
}
```

Response examples:

Example 1: An error code is returned.

```
{
  "success": false,
  "data": null,
  "failCode": 20605,
  "message": "The time cannot be a negative number."
}
```

Example 2: The plant list data is returned.

```
{
  "success": true,
```

```
"data": {
  "list": [
    {
      "plantCode": "NE=12345678",
      "plantName": "NMplant1",
      "plantAddress": null,
      "longitude": null,
      "latitude": null,
      "capacity": 146.5,
      "contactPerson": "",
      "contactMethod": "",
      "gridConnectionDate": "2022-11-21T16:23:00+08:00"
    },
    {
      "plantCode": "NE=23456789",
      "plantName": "plant2",
      "plantAddress": null,
      "longitude": null,
      "latitude": null,
      "capacity": 123.3,
      "contactPerson": "",
      "contactMethod": "",
      "gridConnectionDate": "2022-11-21T16:30:28-12:00"
    }
  ],
  "pageCount": 1,
  "pageNo": 1,
  "pageSize": 100,
  "total": 2
},
"failCode": 0,
"message": "get plant list success"
}
```

5.1.1.2 Device List API

API Description

This API is used to obtain basic device information. Before invoking other APIs to obtain device data, you need to call this API to obtain the device ID.

When you query devices by plant ID set, devices of a maximum of 100 plants can be queried at a time.



CAUTION

If no data is queried or only some data is returned when the device list API is used, rectify the fault by referring to [8.3 Why Is No Data or Only Part of Data Found When I Call a Northbound API for Data Query?](#)

Request URL

`https://Domain name of the management system/thirdData/getDevList`

Request Mode

HTTP method: POST

Request Parameters

Parameter	Description	Data Type	Mandatory/Optional
stationCodes	Plant ID list. Multiple plant IDs are separated by commas (,). The plant IDs are obtained from plantCode in 5.1.1.1 Plant List API .	String	Mandatory

Response Packets

Parameter		Description	Data Type	Remarks
success		Request success or failure flag. true : The request succeeded. false : The request failed.	Boolean	Request success or failure flag.
failCode		Error code. Value 0 indicates that the status is normal. For other error codes, see 6.1 Error Code List .	Integer	-
params	Parameters	-	-	-
	stationCodes	Plant ID list in the request parameter.	String	-
	currentTime	Current system timestamp, in milliseconds.	Long	Example : 1501862400000
message		Optional message.	String	-
data	Parameters	Returned data. The data contains the object parameter list of each device.	List	-
	id	Device ID.	Long	-
	devDn	Unique device ID in the system.	String	-
	devName	Device name.	String	-
	stationCode	Plant ID.	String	-

Parameter		Description	Data Type	Remarks
	esnCode	Device SN.	String	-
	devTypeId	Device type ID. The following device types are supported: 1: string inverter 2: SmartLogger 8: STS 10: EMI 13: protocol converter 16: general device 17: grid meter 22: PID 37: Pinnet data logger 38: residential inverter 39: battery 40: backup box 41: ESS 45: PLC 46: optimizer 47: power sensor 62: Dongle 63: distributed SmartLogger 70: safety box 60001: mains 60003: genset 60043: SSU group 60044: SSU 60092: power converter 60014: lithium battery rack 60010: AC output power distribution 23070: EMMA	Integer	If devTypeId that is not described in this document is returned, see 8.10 devTypeId Reference Table Returned by the Device List API .

Parameter		Description	Data Type	Remarks
	model	Device type.	String	This field is returned only for the device types mentioned in the devTypeId field description.
	softwareVersion	Software version.	String	-
	optimizerNumber	Quantity of optimizers.	Integer	
	invType	Inverter model (only applicable to inverters).	String	-
	longitude	Longitude.	Double	-
	latitude	Latitude.	Double	

Examples

Request example:

```
{
  "stationCodes": "NE=33554875,NE=33554876"
}
```

Response examples:

Example 1: An error code is returned.

```
{
  "success": false,
  "data": null,
  "failCode": 20009,
  "params": {
    "stationCodes": "NE=33554875,NE=33554876",
    "currentTime": 1503046597854
  },
  "message": null
}
```

Example 2: The device list is returned.

```
{
  "success": true,
  "data": [
    {
      "id": -214543629611879,
      "devDn": "NE=45112560",

```

```
{
  "devName": "5fbfk4",
  "stationCode": "NE=33554876",
  "esnCode": "5fbfk4",
  "devTypeId": 1,
  "softwareVersion": "V100R001PC666",
  "invType": "SUN2000-17KTL",
  "longitude": null,
  "latitude": null,
  "model": "SUN2000-17KTL"
},
{
  "id": -214091680973855,
  "devDn": "NE=4511256",
  "devName": "6fbfk11",
  "stationCode": "NE=33554876",
  "esnCode": "6fbfk11",
  "devTypeId": 1,
  "softwareVersion": "V100R001PC666",
  "invType": "SUN2000-17KTL",
  "longitude": null,
  "latitude": null,
  "model": "SUN2000-17KTL"
}
],
"failCode": 0,
"params": {
  "stationCodes": "NE=33554875,NE=33554876",
  "currentTime": 1503046597854
},
"message": null
}
```

NOTE

Prerequisites for obtaining data: The API account has the permission to access this API.

Request example:

The screenshot displays a REST client interface for a POST request to the endpoint `/thirdData/getDevList`. The request headers include `Content-Type: application/json; charset=UTF-8` and `XSRF-TOKEN: x-mko48b...jqlrusalgnz7qz`. The request body is a JSON object with `"stationCodes": "NE=33554875,NE=33554876"`. The response is a 200 OK status with a JSON body containing an array of device data.

Request:

```
{
  "stationCodes": "NE=33554875,NE=33554876"
}
```

Response:

```
{
  "data": [
    {
      "devName": "te:...",
      "devTypeId": 38,
      "esnCode": "te:..."
    }
  ]
}
```

Request Headers:

- Accept: */*
- Accept-Encoding: gzip, deflate, br
- Accept-Language: zh-CN,zh;q=0.9
- Connection: keep-alive
- Content-Length: 48
- Content-Type: application/json; charset=UTF-8
- Cookies: XSRF-TOKEN=x-mko48b...jqlrusalgnz7qz
- Host: ...
- Origin: chrome-extension://...
- Sec-Fetch-Dest: empty
- Sec-Fetch-Mode: cors
- Sec-Fetch-Site: none
- User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/87.0.4280.88 Safari/537.36
- XSRF-TOKEN: x-mko48b...jqlrusalgnz7qz

Request Payload:

```
{
  "stationCodes": "NE=33554875,NE=33554876"
}
```

Prerequisites for obtaining data: The API account has the permission to access this API.
Request example:

The screenshot displays a REST client interface with the following details:

- Request:**
 - Method:** POST
 - URL:** https://localhost:3000/api/data
 - Headers:**
 - Content-Type: application/json
 - XSRF-TOKEN: x-mkcdll...
 - Body:**

```
{
  "devId": "100...",
  "devType": "30"
}
```
- Response:**
 - Status:** 200 OK
 - Body:**

```
{
  "data": [
    {
      "devId": "100...",
      "dataTestMap": {
        "pv1_u": 200.0,
        "pv2_u": 200.0, ...
      }
    }
  ]
}
```
- Response Headers:**
 - Access-Control-Allow-Origin: *
 - Access-Encoding: gzip, deflate, br
 - Access-Language: zh-CN, zh;q=0.9
 - Connection: keep-alive
 - Content-Length: 51
 - Content-Type: application/json
 - Cookies: XSRF-TOKEN=x-mkcdll...; user_time_show_dst=1; locale=zh-cn; session=deleted; 3E552D02=3E444C4B0F06FCE376F5E8B08
 - Host: localhost:3000
 - Origin: chrome-extension://efojaggenchagidmlietcmf
 - Sec-Fetch-Dest: empty
 - Sec-Fetch-Mode: cors
 - Sec-Fetch-Site: none
 - User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/87.0.4380.80 Safari/537.36

API Description

For details about the data list that can be queried through this API, see the historical device data list below.

<https://Domain name of the management system/thirdData/getDevHistoryKpi>

HTTP method: POST

Request Parameters

Parameter	Description	Data Type	Mandatory/Optional
devIds	List of device IDs. The device IDs are obtained from id in 5.1.1.2 Device List API . Use commas (,) to separate multiple device IDs. Either sns or devIds must be set.	String	Optional
sns	Device SN list. Multiple device SNs are separated by commas (,). Either sns or devIds must be set.	String	Optional
devTypeId	Device type ID. The values of devTypeId obtained in 5.1.1.2 Device List API are used. The following device types are supported: 1: string inverter 10: EMI 17: grid meter 38: residential inverter 39: battery 41: ESS 47: power sensor 60001: mains 60003: genset 60043: SSU group 60014: lithium battery rack 60010: AC output power distribution	Integer	Mandatory
startTime	Start timestamp (in milliseconds), for example, 1501862400000.	Long	Mandatory
endTime	End timestamp (in milliseconds), for example, 1501862400000.	Long	Mandatory

Response Packet

Parameter		Description	Data Type	Remarks
success		Request success or failure flag. true: The request succeeded. false: The request failed.	Boolean	Request success or failure flag
failCode		Error code. Value 0 indicates that the status is normal. For other error codes, see 6.1 Error Code List .	Integer	-
parameters	Parameters	-	-	-
	devIds	Device ID list in the request parameter.	String	-
	sns	Device SN list in the request parameter.	String	-
	devTypeId	Device type ID in the request parameter.	Integer	-
	startTime	Start timestamp, in milliseconds.	Long	Example : 1501862400000
	endTime	End timestamp, in milliseconds.	Long	Example : 1501862400000
	currentTime	Current system timestamp, in milliseconds.	Long	Example : 1501862400000
message		Optional message.	String	-
data	Parameters	Returned data. The data contains the 5-minute data object list of each device.	List	5-minute data of a device in a day
	devId	Device ID.	Long	-
	sn	Device SN.	String	

Parameter		Description	Data Type	Remarks
	collectTime	Collection time in milliseconds in the request parameter.	Long	-
	dataItemMap	Content of each data item, which is returned in the key-value format. The data item content varies depending on the device type. For details about the data item list, see the historical device data list below.	Map	5-minute device data

Historical Device Data List

Device Type	Key	Name	Unit	Return Value Type	Remarks
ID: 1 String inverter	inverter_state	Inverter state. For details, see the following: Table 5-2	None	Double	
	ab_u	A-B line voltage of grid	V	Double	
	bc_u	B-C line voltage of grid	V	Double	
	ca_u	C-A line voltage of grid	V	Double	
	a_u	Phase A voltage	V	Double	
	b_u	Phase B voltage	V	Double	

Device Type	Key	Name	Unit	Return Value Type	Remarks
	c_u	Phase C voltage	V	Double	
	a_i	Phase A current of grid	A	Double	
	b_i	Phase B current of grid	A	Double	
	c_i	Phase C current of grid	A	Double	
	efficiency	Inverter conversion efficiency (manufacturer)	%	Double	
	temperature	Internal temperature	°C	Double	
	power_factor	Power factor	None	Double	
	elec_freq	Grid frequency	Hz	Double	
	active_power	Active power	kW	Double	
	reactive_power	Output reactive power	kVar	Double	
	day_cap	Energy yield of current day.	kWh	Double	
	mppt_power	MPPT total input power	kW	Double	

Device Type	Key	Name	Unit	Return Value Type	Remarks
	pv1_u	PV1 input voltage	V	Double	
	pv2_u	PV2 input voltage	V	Double	
	pv3_u	PV3 input voltage	V	Double	
	pv4_u	PV4 input voltage	V	Double	
	pv5_u	PV5 input voltage	V	Double	
	pv6_u	PV6 input voltage	V	Double	
	pv7_u	PV7 input voltage	V	Double	
	pv8_u	PV8 input voltage	V	Double	
	pv9_u	PV9 input voltage	V	Double	
	pv10_u	PV10 input voltage	V	Double	
	pv11_u	PV11 input voltage	V	Double	
	pv12_u	PV12 input voltage	V	Double	

Device Type	Key	Name	Unit	Return Value Type	Remarks
	pv13_u	PV13 input voltage	V	Double	
	pv14_u	PV14 input voltage	V	Double	
	pv15_u	PV15 input voltage	V	Double	
	pv16_u	PV16 input voltage	V	Double	
	pv17_u	PV17 input voltage	V	Double	
	pv18_u	PV18 input voltage	V	Double	
	pv19_u	PV19 input voltage	V	Double	
	pv20_u	PV20 input voltage	V	Double	
	pv21_u	PV21 input voltage	V	Double	
	pv22_u	PV22 input voltage	V	Double	
	pv23_u	PV23 input voltage	V	Double	
	pv24_u	PV24 input voltage	V	Double	

Device Type	Key	Name	Unit	Return Value Type	Remarks
	pv25_u	PV25 input voltage	V	Double	
	pv26_u	PV26 input voltage	V	Double	
	pv27_u	PV27 input voltage	V	Double	
	pv28_u	PV28 input voltage	V	Double	
	pv29_u	PV29 input voltage	V	Double	
	pv30_u	PV30 input voltage	V	Double	
	pv31_u	PV31 input voltage	V	Double	
	pv32_u	PV32 input voltage	V	Double	
	pv33_u	PV33 input voltage	V	Double	
	pv34_u	PV34 input voltage	V	Double	
	pv35_u	PV35 input voltage	V	Double	
	pv36_u	PV36 input voltage	V	Double	

Device Type	Key	Name	Unit	Return Value Type	Remarks
	pv1_i	PV1 input current	A	Double	
	pv2_i	PV2 input current	A	Double	
	pv3_i	PV3 input current	A	Double	
	pv4_i	PV4 input current	A	Double	
	pv5_i	PV5 input current	A	Double	
	pv6_i	PV6 input current	A	Double	
	pv7_i	PV7 input current	A	Double	
	pv8_i	PV8 input current	A	Double	
	pv9_i	PV9 input current	A	Double	
	pv10_i	PV10 input current	A	Double	
	pv11_i	PV11 input current	A	Double	
	pv12_i	PV12 input current	A	Double	

Device Type	Key	Name	Unit	Return Value Type	Remarks
	pv13_i	PV13 input current	A	Double	
	pv14_i	PV14 input current	A	Double	
	pv15_i	PV15 input current	A	Double	
	pv16_i	PV16 input current	A	Double	
	pv17_i	PV17 input current	A	Double	
	pv18_i	PV18 input current	A	Double	
	pv19_i	PV19 input current	A	Double	
	pv20_i	PV20 input current	A	Double	
	pv21_i	PV21 input current	A	Double	
	pv22_i	PV22 input current	A	Double	
	pv23_i	PV23 input current	A	Double	
	pv24_i	PV24 input current	A	Double	

Device Type	Key	Name	Unit	Return Value Type	Remarks
	pv25_i	PV25 input current	A	Double	
	pv26_i	PV26 input current	A	Double	
	pv27_i	PV27 input current	A	Double	
	pv28_i	PV28 input current	A	Double	
	pv29_i	PV29 input current	A	Double	
	pv30_i	PV30 input current	A	Double	
	pv31_i	PV31 input current	A	Double	
	pv32_i	PV32 input current	A	Double	
	pv33_i	PV33 input current	A	Double	
	pv34_i	PV34 input current	A	Double	
	pv35_i	PV35 input current	A	Double	
	pv36_i	PV36 input current	A	Double	

Device Type	Key	Name	Unit	Return Value Type	Remarks
	total_cap	Total yield	kWh	Double	
	open_time	Inverter startup time	ms	Double	
	close_time	Inverter shutdown time	ms	Double	
	mppt_total_cap	Total DC input energy	kWh	Double	
	mppt_1_cap	MPPT 1 DC total yield	kWh	Double	
	mppt_2_cap	MPPT 2 DC total yield	kWh	Double	
	mppt_3_cap	MPPT 3 DC total yield	kWh	Double	
	mppt_4_cap	MPPT 4 DC total yield	kWh	Double	
	mppt_5_cap	MPPT 5 DC total yield	kWh	Double	
	mppt_6_cap	MPPT 6 DC total yield	kWh	Double	
	mppt_7_cap	MPPT 7 DC total yield	kWh	Double	
	mppt_8_cap	MPPT 8 DC total yield	kWh	Double	

Device Type	Key	Name	Unit	Return Value Type	Remarks
	mppt_9_cap	MPPT 9 DC total yield	kWh	Double	
	mppt_10_cap	MPPT 10 DC total yield	kWh	Double	
ID: 38 Residential inverter	inverter_state	Inverter state. For details, see Table 5-2 .	None	Double	
	ab_u	A-B line voltage of grid	V	Double	
	bc_u	B-C line voltage of grid	V	Double	
	ca_u	C-A line voltage of grid	V	Double	
	a_u	Phase A voltage	V	Double	
	b_u	Phase B voltage	V	Double	
	c_u	Phase C voltage	V	Double	
	a_i	Phase A current of grid	A	Double	
	b_i	Phase B current of grid	A	Double	
	c_i	Phase C current of grid	A	Double	

Device Type	Key	Name	Unit	Return Value Type	Remarks
	efficiency	Inverter conversion efficiency (manufacturer)	%	Double	
	temperature	Internal temperature	°C	Double	
	power_factor	Power factor	None	Double	
	elec_freq	Grid frequency	Hz	Double	
	active_power	Active power	kW	Double	
	reactive_power	Output reactive power	kVar	Double	
	day_cap	Energy yield of current day.	kWh	Double	
	mppt_power	MPPT total input power	kW	Double	
	pv1_u	PV1 input voltage	V	Double	
	pv2_u	PV2 input voltage	V	Double	
	pv3_u	PV3 input voltage	V	Double	
	pv4_u	PV4 input voltage	V	Double	

Device Type	Key	Name	Unit	Return Value Type	Remarks
	pv5_u	PV5 input voltage	V	Double	
	pv6_u	PV6 input voltage	V	Double	
	pv7_u	PV7 input voltage	V	Double	
	pv8_u	PV8 input voltage	V	Double	
	pv1_i	PV1 input current	A	Double	
	pv2_i	PV2 input current	A	Double	
	pv3_i	PV3 input current	A	Double	
	pv4_i	PV4 input current	A	Double	
	pv5_i	PV5 input current	A	Double	
	pv6_i	PV6 input current	A	Double	
	pv7_i	PV7 input current	A	Double	
	pv8_i	PV8 input current	A	Double	

Device Type	Key	Name	Unit	Return Value Type	Remarks
	total_cap	Total yield	kWh	Double	
	open_time	Inverter startup time	ms	Double	
	close_time	Inverter shutdown time	ms	Double	
	mppt_1_cap	MPPT 1 DC total yield	kWh	Double	
	mppt_2_cap	MPPT 2 DC total yield	kWh	Double	
	mppt_3_cap	MPPT 3 DC total yield	kWh	Double	
	mppt_4_cap	MPPT 4 DC total yield	kWh	Double	
ID: 10 EMI	temperature	Temperature	°C	Double	
	pv_temperature	PV temperature	°C	Double	
	wind_speed	Wind speed	m/s	Double	
	wind_direction	Wind direction	Degree	Double	
	radiant_total	Daily irradiation	MJ/m ²	Double	

Device Type	Key	Name	Unit	Return Value Type	Remarks
	radiant_line	Irradiance	W/m ²	Double	
	horiz_radiant_line	Horizontal irradiance	W/m ²	Double	Invalid field. The default value is null .
	horiz_radiant_total	Horizontal irradiation	MJ/m ²	Double	Invalid field. The default value is null .
ID: 17 Grid meter	ab_u	A-B line voltage of grid	V	Double	
	bc_u	B-C line voltage of grid	V	Double	
	ca_u	C-A line voltage of grid	V	Double	
	a_u	Phase A voltage (AC output)	V	Double	
	b_u	Phase B voltage (AC output)	V	Double	
	c_u	Phase C voltage (AC output)	V	Double	
	a_i	Phase A current of grid (IA)	A	Double	
	b_i	Phase B current of grid (IB)	A	Double	

Device Type	Key	Name	Unit	Return Value Type	Remarks
	c_i	Phase C current of grid (IC)	A	Double	
	active_power	Active power	W	Double	
	power_factor	Power factor	None	Double	
	active_cap	Active energy (positive active energy)	kWh	Double	
	reactive_power	Reactive power	kVar	Double	
	reverse_active_cap	Negative active energy	kWh	Double	
	forward_reactive_cap	Positive reactive energy	kWh	Double	
	reverse_reactive_cap	Negative reactive energy	kWh	Double	
	active_power_a	Active power PA	kW	Double	
	active_power_b	Active power PB	kW	Double	
	active_power_c	Active power PC	kW	Double	
	reactive_power_a	Reactive power QA	kVar	Double	Invalid field. The default value is null .

Device Type	Key	Name	Unit	Return Value Type	Remarks
	reactive_power_b	Reactive power QB	kVar	Double	Invalid field. The default value is null .
	reactive_power_c	Reactive power QC	kVar	Double	Invalid field. The default value is null .
	total_apparent_power	Total apparent power	kVA	Double	
	grid_frequency	Grid frequency	Hz	Double	Invalid field. The default value is null .
	reverse_active_peak	Negative active energy (peak)	kWh	Double	Invalid field. The default value is null .
	reverse_active_power	Negative active energy (shoulder)	kWh	Double	Invalid field. The default value is null .
	reverse_active_valley	Negative active energy (off-peak)	kWh	Double	Invalid field. The default value is null .
	reverse_active_top	Negative active energy (sharp)	kWh	Double	Invalid field. The default value is null .

Device Type	Key	Name	Unit	Return Value Type	Remarks
	positive_active_peak	Positive active energy (peak)	kWh	Double	Invalid field. The default value is null .
	positive_active_power	Positive active energy (shoulder)	kWh	Double	Invalid field. The default value is null .
	positive_active_valley	Positive active energy (off-peak)	kWh	Double	Invalid field. The default value is null .
	positive_active_top	Positive active energy (sharp)	kWh	Double	Invalid field. The default value is null .
	reverse_reactive_peak	Negative reactive energy (peak)	kVar	Double	Invalid field. The default value is null .
	reverse_reactive_power	Negative reactive energy (shoulder)	kVar	Double	Invalid field. The default value is null .
	reverse_reactive_valley	Negative reactive energy (off-peak)	kVar	Double	Invalid field. The default value is null .
	reverse_reactive_top	Negative reactive energy (sharp)	kVar	Double	Invalid field. The default value is null .

Device Type	Key	Name	Unit	Return Value Type	Remarks
	positive_reactive_peak	Positive reactive energy (peak)	kVar	Double	Invalid field. The default value is null .
	positive_reactive_power	Positive reactive energy (shoulder)	kVar	Double	Invalid field. The default value is null .
	positive_reactive_valley	Positive reactive energy (off-peak)	kVar	Double	Invalid field. The default value is null .
	positive_reactive_top	Positive reactive energy (sharp)	kVar	Double	Invalid field. The default value is null .
ID: 47 Power sensor	meter_status	Meter state (0: offline; 1: normal)	None	Double	
	meter_u	Grid voltage	V	Double	
	meter_i	Grid current	A	Double	
	active_power	Active power	W	Double	
	reactive_power	Reactive power	Var	Double	
	power_factor	Power factor	None	Double	

Device Type	Key	Name	Unit	Return Value Type	Remarks
	grid_frequency	Grid frequency	Hz	Double	
	active_cap	Active energy (positive active energy)	kWh	Double	
	reverse_active_cap	Negative active energy	kWh	Double	
	ab_u	A-B line voltage of grid	V	Double	
	bc_u	B-C line voltage of grid	V	Double	
	ca_u	C-A line voltage of grid	V	Double	
	b_u	Phase B voltage (AC output)	V	Double	
	c_u	Phase C voltage (AC output)	V	Double	
	b_i	Phase B current of grid (IB)	A	Double	
	c_i	Phase C current of grid (IC)	A	Double	
	forward_reactive_cap	Positive reactive energy	kWh	Double	Invalid field. The default value is null .

Device Type	Key	Name	Unit	Return Value Type	Remarks
	reverse_reactive_cap	Negative reactive energy	kWh	Double	Invalid field. The default value is null .
	active_power_a	Active power PA	kW	Double	
	active_power_b	Active power PB	kW	Double	
	active_power_c	Active power PC	kW	Double	
	reactive_power_a	Reactive power QA	kVar	Double	Invalid field. The default value is null .
	reactive_power_b	Reactive power QB	kVar	Double	Invalid field. The default value is null .
	reactive_power_c	Reactive power QC	kVar	Double	Invalid field. The default value is null .
	total_apparent_power	Total apparent power	kVA	Double	Invalid field. The default value is null .
	reverse_active_peak	Negative active energy (peak)	kWh	Double	Invalid field. The default value is null .

Device Type	Key	Name	Unit	Return Value Type	Remarks
	reverse_active_power	Negative active energy (shoulder)	kWh	Double	Invalid field. The default value is null .
	reverse_active_valley	Negative active energy (off-peak)	kWh	Double	Invalid field. The default value is null .
	reverse_active_top	Negative active energy (sharp)	kWh	Double	Invalid field. The default value is null .
	positive_active_peak	Positive active energy (peak)	kWh	Double	Invalid field. The default value is null .
	positive_active_power	Positive active energy (shoulder)	kWh	Double	Invalid field. The default value is null .
	positive_active_valley	Positive active energy (off-peak)	kWh	Double	Invalid field. The default value is null .
	positive_active_top	Positive active energy (sharp)	kWh	Double	Invalid field. The default value is null .
	reverse_reactive_peak	Negative reactive energy (peak)	kVar	Double	Invalid field. The default value is null .

Device Type	Key	Name	Unit	Return Value Type	Remarks
	reverse_reactive_power	Negative reactive energy (shoulder)	kVar	Double	Invalid field. The default value is null .
	reverse_reactive_valley	Negative reactive energy (off-peak)	kVar	Double	Invalid field. The default value is null .
	reverse_reactive_top	Negative reactive energy (sharp)	kVar	Double	Invalid field. The default value is null .
	positive_reactive_peak	Positive reactive energy (peak)	kVar	Double	Invalid field. The default value is null .
	positive_reactive_power	Positive reactive energy (shoulder)	kVar	Double	Invalid field. The default value is null .
	positive_reactive_valley	Positive reactive energy (off-peak)	kVar	Double	Invalid field. The default value is null .
	positive_reactive_top	Positive reactive energy (sharp)	kVar	Double	Invalid field. The default value is null .
ID: 39 Residential battery	battery_status	Battery running state (0: offline; 1: standby; 2: running; 3: faulty; 4: hibernating)	None	Double	

Device Type	Key	Name	Unit	Return Value Type	Remarks
	max_charge_power	Maximum charge power	W	Double	
	max_discharge_power	Maximum discharge power	W	Double	
	ch_discharge_power	Charge/ Discharge power	W	Double	
	busbar_u	Battery voltage	V	Double	
	battery_soc	Battery SOC	%	Double	
	battery_soh	Battery SOH	None	Double	
	ch_discharge_model	Charge/ Discharge mode (0: none; 1: forced charge/ discharge; 2: time-of-use price; 3: fixed charge/ discharge; 4: automatic charge/ discharge)	None	Double	
	charge_cap	Energy charged today	kWh	Double	
	discharge_cap	Energy discharged today	kWh	Double	
ID: 41 C&I and utility ESS	ch_discharge_power	Charge/ Discharge power	W	Double	

Device Type	Key	Name	Unit	Return Value Type	Remarks
	battery_soc	Battery SOC	%	Double	
	charge_cap	Energy charged today	kWh	Double	
	discharge_cap	Energy discharged today	kWh	Double	
ID: 60001 Mains (supported only in the Power-M scenario)	mains_state	Mains status (0: mains unavailable; 1: mains available)	None	Double	
	ac_voltage	AC voltage	V	Double	
	ac_current	AC current	A	Double	
	active_power	Active power	kW	Double	
	ac_frequency	AC frequency	Hz	Double	
ID: 60003 Genset (supported only in the Power-M scenario)	load_rate	Load rate	%	Double	
	running_state	Running status (0: unknown; 1: stopped; 2: running)	None	Double	
ID: 60043 SSU group (supported only in the Power-M scenario)	total_output_current	Total output current	A	Double	
	total_output_energy	Total output energy	kWh	Double	

Device Type	Key	Name	Unit	Return Value Type	Remarks
ID: 60014 Lithium battery rack (supported only in the Power-M scenario)	soc	SOC	%	Double	
	voltage	Voltage	V	Double	
	average_temperature	Average temperature	°C	Double	
ID: 60010 AC output power distribution (supported only in the Power-M scenario)	ac_frequency	AC frequency	Hz	Double	
	total_load_active_power	Total load active power	kW	Double	

Table 5-2 Inverter state (inverter_state) description

State Value	Description
0	Standby: initializing
1	Standby: insulation resistance detecting
2	Standby: irradiation detecting
3	Standby: grid detecting
256	Start
512	Grid-connected
513	Grid-connected: power limited
514	Grid-connected: self-derating
768	Shutdown: on fault
769	Shutdown: on command
770	Shutdown: OVGR
771	Shutdown: communication interrupted

State Value	Description
772	Shutdown: power limited
773	Shutdown: manual startup required
774	Shutdown: DC switch disconnected
1025	Grid scheduling: $\cos\psi$ -P curve
1026	Grid scheduling: Q-U curve
1280	Ready for terminal test
1281	Terminal testing...
1536	Inspection in progress
1792	AFCI self-check
2048	I-V scanning
2304	DC input detection
40960	Standby: no irradiation
45056	Communication interrupted (written by SmartLogger)
49152	Loading... (written by SmartLogger)

Examples

Request example:

```
{
  "devIds": "214060404588862,213472461631079",
  "devTypeId": 1,
  "startTime": 1501862400000,
  "endTime": 1501872400000
}
```

Response examples:

Example 1: An error code is returned.

```
{
  "success": false,
  "data": null,
  "failCode": 20009,
  "params": {
    "devIds": "214060404588862,213472461631079",
    "devTypeId": 1,
    "startTime": 1501862400000,
    "endTime": 1501872400000,
    "currentTime": 1503046597854
  },
  "message": null
}
```

Example 2: 5-minute device data is returned.

```
{
  "success": true,
```

```
"data":[
  {
    "dataItemMap":{
      "pv7_u":null,
      "pv1_u":575.3,
      "b_u":286.1,
      "c_u":286.9,
      "pv6_u":576.1,
      "temperature":44.6,
      "open_time":null,
      "b_i":24.9,
      "bc_u":495.6,
      "pv9_u":null,
      "pv8_u":null,
      "c_i":25,
      "mppt_total_cap":null,
      "pv9_i":null,
      "mppt_3_cap":null,
      "mppt_2_cap":null,
      "inverter_state":512,
      "pv8_i":null,
      "mppt_1_cap":null,
      "pv6_i":7.1,
      "mppt_power":21.962,
      "pv1_i":7.1,
      "total_cap":655.37,
      "ab_u":495.4,
      "pv7_i":null,
      "pv13_u":null,
      "reactive_power":20.95,
      "pv10_u":null,
      "pv12_i":null,
      "pv11_i":null,
      "pv3_i":7.1,
      "pv11_u":null,
      "pv2_i":7.1,
      "pv13_i":null,
      "power_factor":0,
      "pv12_u":null,
      "pv5_i":7.2,
      "active_power":21.05,
      "elec_freq":50.05,
      "pv10_i":null,
      "pv4_i":7,
      "mppt_4_cap":null,
      "mppt_5_cap":0,
      "mppt_6_cap":0,
      "mppt_7_cap":0,
      "mppt_8_cap":0,
      "mppt_9_cap":0,
      "mppt_10_cap":0,
      "pv4_u":577.8,
      "close_time":null,
      "day_cap":159.26,
      "ca_u":496.9,
      "a_i":24.9,
      "pv5_u":576.1,
      "a_u":286,
      "pv3_u":577.8,
      "pv14_u":null,
      "pv14_i":null,
      "pv15_u":0,
      "pv15_i":0,
      "pv16_u":0,
      "pv16_i":0,
      "pv17_u":0,
      "pv17_i":0,
      "pv18_u":0,
      "pv18_i":0,
    }
  }
]
```

```
"pv19_u":0,
"pv19_i":0,
"pv20_u":0,
"pv20_i":0,
"pv21_u":0,
"pv21_i":0,
"pv22_u":0,
"pv22_i":0,
"pv23_u":0,
"pv23_i":0,
"pv24_u":0,
"pv24_i":0,
"pv25_u":0,
"pv25_i":0,
"pv26_u":0,
"pv26_i":0,
"pv27_u":0,
"pv27_i":0,
"pv28_u":0,
"pv28_i":0,
"efficiency":null,
"pv2_u":575.3
},
"devId":213472461631079,
"collectTime":1501862400000
},
{
  "dataItemMap":{
    "pv7_u":null,
    "pv1_u":575.3,
    "b_u":286.1,
    "c_u":286.9,
    "pv6_u":576.1,
    "temperature":44.6,
    "open_time":null,
    "b_i":24.9,
    "bc_u":495.6,
    "pv9_u":null,
    "pv8_u":null,
    "c_i":25,
    "mppt_total_cap":null,
    "pv9_i":null,
    "mppt_3_cap":null,
    "mppt_2_cap":null,
    "inverter_state":512,
    "pv8_i":null,
    "mppt_1_cap":null,
    "pv6_i":7.1,
    "mppt_power":21.962,
    "pv1_i":7.1,
    "total_cap":655.37,
    "ab_u":495.4,
    "pv7_i":null,
    "pv13_u":null,
    "reactive_power":20.95,
    "pv10_u":null,
    "pv12_i":null,
    "pv11_i":null,
    "pv3_i":7.1,
    "pv11_u":null,
    "pv2_i":7.1,
    "pv13_i":null,
    "power_factor":0,
    "pv12_u":null,
    "pv5_i":7.2,
    "active_power":21.05,
    "elec_freq":50.05,
    "pv10_i":null,
    "pv4_i":7,
```

```
        "mppt_4_cap":null,
        "mppt_5_cap":0,
        "mppt_6_cap":0,
        "mppt_7_cap":0,
        "mppt_8_cap":0,
        "mppt_9_cap":0,
        "mppt_10_cap":0,
        "pv4_u":577.8,
        "close_time":null,
        "day_cap":159.26,
        "ca_u":496.9,
        "a_i":24.9,
        "pv5_u":576.1,
        "a_u":286,
        "pv3_u":577.8,
        "pv14_u":null,
        "pv14_i":null,
        "pv15_u":0,
        "pv15_i":0,
        "pv16_u":0,
        "pv16_i":0,
        "pv17_u":0,
        "pv17_i":0,
        "pv18_u":0,
        "pv18_i":0,
        "pv19_u":0,
        "pv19_i":0,
        "pv20_u":0,
        "pv20_i":0,
        "pv21_u":0,
        "pv21_i":0,
        "pv22_u":0,
        "pv22_i":0,
        "pv23_u":0,
        "pv23_i":0,
        "pv24_u":0,
        "pv24_i":0,
        "pv25_u":0,
        "pv25_i":0,
        "pv26_u":0,
        "pv26_i":0,
        "pv27_u":0,
        "pv27_i":0,
        "pv28_u":0,
        "pv28_i":0,
        "efficiency":null,
        "pv2_u":575.3
    },
    "devId":213472461631079,
    "collectTime":1501862700000
}
],
"failCode":0,
"params":{
    "devIds":["214060404588862,213472461631079",
    "devType":1,
    "startTime":1501862400000,
    "endTime":1501872400000,
    "currentTime":1503046597854
},
"message":null
}
```

NOTE

Prerequisites for obtaining data: The API account has the permission to access this API.

5.1.3 Alarm

6References

6.1 Error Code List

No.	Error Code	Description
1	20001	The unique code of the PV module library configuration is empty.
2	20002	The API account has been disabled.
3	20003	The API account has expired.
4	20004	The server is abnormal.
5	20005	The device ID, SN, or DN cannot be empty.
6	20006	Some devices do not match the device type.
7	20007	The system does not have the desired power plant resources.
8	20008	The value of devTypeId cannot be empty.
9	20010	The plant list cannot be empty.
10	20011	The device list cannot be empty.
11	20012	The query time cannot be empty.
12	20013	The device type is incorrect. The API does not support operations on some devices.
13	20015	A maximum of 100 plants can be queried at a time.
14	20016	A maximum of 100 devices can be queried at a time.
15	20017	A maximum of 100 devices can be queried at a time.

No.	Error Code	Description
16	20019	The switch type is incorrect. 1 and 2 indicate switch-on and switch-off respectively.
17	305	You are not online and need to log in again.
18	401	You do not have the related data API permission. • For details about how to locate the fault in the OAuth Connect scenario, see 8.1.5 Why Does the Northbound API Return Error Code 401? • For details about how to check the API account access scenario, see 8.2.5 Why Does the Northbound API Return Error Code 401?
19	407	The API access frequency is too high.
20	20028	The system does not have related user information.
21	20029	Invalid user information.
22	20030	Failed to create the I-V curve diagnosis task.
23	20034	The task does not exist.
24	20038	In the input parameters, the authorization code list is empty (null), or the number of authorization codes is out of range. The allowed range is [0, 1000].
25	20039	In the input parameters, the DOD value is out of range. The allowed range is [0, 100].
26	20040	The charge/discharge switch parameter value is invalid.
27	20041	The control type cannot be empty for forced charge and discharge.
28	20042	The target SOC for charge/discharge is empty or invalid.
29	20043	The charge/discharge duration is empty or invalid.
30	20044	The unique ID of a charge/discharge task cannot be empty.
31	20045	Unauthorized PV plants exist in the input parameters.
32	20046	Unauthorized devices exist in the input parameters.
33	20047	The forced charge/discharge power in the input parameters is invalid.
34	20048	Duplicate charge and discharge task ID.
35	20049	Failed to deliver the charging and discharging task.
36	20050	The charging and discharging task query parameter does not exist.

No.	Error Code	Description
37	20051	Failed to set the battery DOD.
38	20055	The plant list and device list parameters cannot be empty at the same time.
39	20056	Resources that are not authorized by the owner exist.
40	20116	The inverter control parameter is incorrect. (The number of PV plants is greater than 100 or equal to 0, or the total number of inverters in the PV plants is greater than 200 or equal to 0.)
41	20117	The plant DN is invalid or the control type is not supported.
42	20200	The system is busy. Try again later.
43	20400	• The API account name or password is incorrect. • The parameter name is incorrect. • The API account is locked. • The password has expired. • The number of online sessions reaches the upper limit. For details, see 8.2.3 Why Does the Northbound API Login Fail?
44	20604	The time parameter is incorrect. The start time cannot be later than or equal to the end time.
45	20605	The time parameter is incorrect. The time parameter contains a negative value.
46	20606	If only the start time is entered and the end time is empty, the start time cannot be later than or equal to the current time.
47	20607	The task list is empty.
48	20608	Duplicate plant IDs exist in the input parameters.
49	20609	The plant networking is abnormal.
50	20610	The plant does not support default configuration.
51	20611	The values of input parameters exceed the valid range.
52	20612	The value of an input parameter is empty.
53	20613	The default setting task fails to be sent for all plants.
54	20614	The network communication is abnormal.
55	20615	The same task is being executed in the current plant.
56	20616	The task ID is empty.

No.	Error Code	Description
57	20617	The query result of the default plant setting task is empty.
58	20618	The number of API calls has reached the maximum number per API account per day.
59	20619	The number of task IDs exceeds 100.
60	20620	The task ID does not exist.
61	20621	The resource is not authorized by the owner.
62	20622	The plant DN is invalid or the account does not have permission on the plant.
63	20623	The plant DN does not belong to the company or subsidiary of the owner.
64	20624	The owner has been bound to the plant.
65	20625	It is not allowed to bind a plant whose sharing request has been accepted by the owner.
66	20626	Failed to bind the plant due to an internal error.
67	20627	The third-party dispatch mode has never been set for the plant through the northbound API. You need to set the working mode to thirdPartyDispatch using 5.2.3 API for Delivering a Task for Setting the Battery Working Mode .
68	20628	The battery working mode is not thirdPartyDispatch .
69	20629	The device in the plant does not support the current setting or query.
70	20631	The device does not exist in the plant.
71	21000	The basic information for the plant creation is empty.
72	21001	The plant name is empty or in an incorrect format.
73	21002	The plant type is empty or incorrect.
74	21003	The grid connection time must be a positive number.
75	21004	The format of the contact name is incorrect.
76	21005	The format of the contact information is incorrect.
77	21006	The C&I plant and utility plant cannot be EV-charger-only plants.
78	21007	The grid connection time cannot be set for an EV-charger-only plant.
79	21008	The string capacity cannot be set for an EV-charger-only plant.

No.	Error Code	Description
80	21009	The electricity price cannot be set for an EV-charger-only plant.
81	21010	>pureChange can only be set to 0 or 1 .
82	21011	The plant name already exists.
83	22000	No related data of connected devices was found.
84	22001	The device registration code is empty.
85	22002	The device is an unauthorized device.
86	22003	The device registration code is incorrect.
87	22004	You will be locked out for 5 minutes due to five consecutive incorrect registration codes.
88	22005	Devices other than chargers are connected to an EV-charger-only plant.
89	22006	The device SN is empty.
90	22007	The device has been bound to another plant.
91	22009	iSitePower-M cannot be connected.
92	22010	The plant with the iSitePower-M cannot connect to other devices.
93	23000	The plant-level string capacity and PV-level string capacity cannot be empty at the same time.
94	23001	The format of the plant string capacity is incorrect.
95	23002	The inverter SN does not exist.
96	23003	The number of inverter SNs is incorrect.
97	23005	The inverter PV string capacity is incorrectly set.
98	23006	The inverter SN or PV string capacity is empty.
99	23007	The quantity in the inverter PV string capacity setting is incorrect.
100	24000	The electricity price must be a positive number.
101	24001	The electricity price date settings must cover a complete year and without overlapping.
102	24002	The electricity price time segments must cover 24 hours of the day without overlapping.
103	24003	The date range is invalid.
104	24004	The time range is invalid.

No.	Error Code	Description
105	24005	The company electricity price is empty.
106	24006	>useCompanyPrice can only be set to 0 or 1 .
107	25000	Additional information configuration is empty.
108	25001	The area code is empty or in an incorrect format.
109	25002	The plant address is empty.
110	25003	The longitude and latitude are empty or in an incorrect format.
111	25004	The safe running time of the plant must be a positive number.
112	25005	The time zone is empty or in an incorrect format.
113	25006	loadStatus can only be set to 0 or 1 .
114	25007	The plant introduction can contain a maximum of 128 characters.
115	26000	Failed to create plants.
116	26001	A maximum of 1000 plants can be created in a day.
117	26002	The company authorized by the API account does not exist.
118	26003	Failed to bind the API account to the device.
119	26004	Failed to unbind the API account from the device.
120	30001	The device ESN list cannot be empty.
121	30002	The ESNs queried at a time cannot exceed 50.
122	30003	The account cannot be empty in the input parameter.
123	30004	The value of pageNo cannot be empty.
124	30005	The value of pageSize cannot be empty.
125	30006	The value of pageSize is out of range. The allowed range is {10, 20, 30, 50, 100}.
126	30007	The values of startTime and endTime must be both provided or empty.
127	30008	Failed to call the internal API.
128	30009	The value of taskName is empty.
129	30010	The value of dns is empty.
130	30011	The value of cleanStatus is empty or invalid.

No.	Error Code	Description
131	30012	The value of environmentalParameters is empty or invalid.
132	30013	The value of modulePanelIrradiance or moduleBackSurfaceTemperature is empty when environmentalParameters is set to 1.
133	30014	The value of scanPointNum must be set to 128.
134	30015	The value of taskId is empty.
135	30016	The value of dn is empty.
136	30017	The value of dns is invalid. The number of devices exceeds 100 or devices on which the API account does not have permission exist.
137	30018	The value of taskName is invalid (for example, null field).
138	30019	The value of moduleBackSurfaceTemperature is out of range. The allowed range is [0.0, 100.0].
139	30020	The value of modulePanelIrradiance is out of range. The allowed range is [600.0, 1500.0].
140	30021	The value of pageNo is smaller than 0.
141	30029	Authentication failed. The entered plant or device parameter is invalid.
142	30030	The input parameter is incorrect.
143	30031	A maximum of 10 devices can be queried at a time.
144	30032	• The time parameter is invalid. The query time segment cannot be longer than three days. • The time-related parameter name is incorrect.
145	30033	The task is in progress.
146	30034	The returned list is empty.
147	30035	The task is to be executed.
148	30036	The task has been canceled.
149	30037	The number of reservation tasks exceeds the maximum (10).
150	30040	The PV string is not configured.
151	30041	The task contains the inverter connected to the optimizer.
152	1	The task is partially successful.
153	2	The task fails.

6.4 Exception Code List

[illegible]

7 Best Practices

7.1 Obtaining a Token

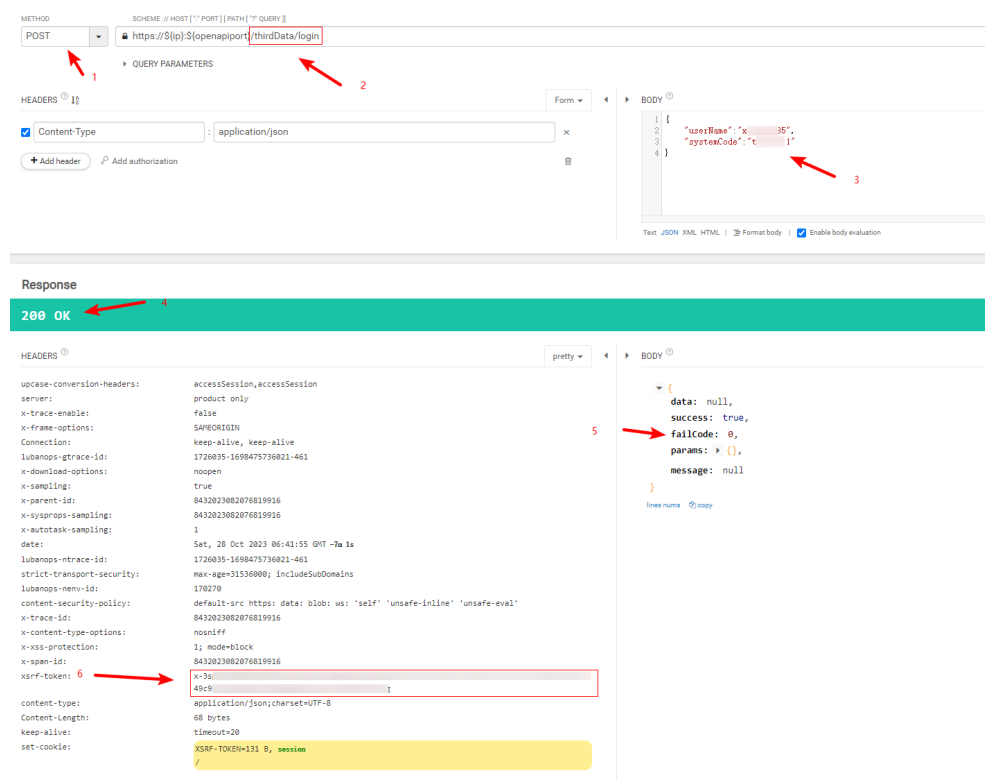
A token is required for the API account created by the administrator to access northbound data. This section describes how to obtain a valid token.

The method and precautions for calling a northbound login API in HTTPS mode are as follows:

1. Method: POST
2. URL: **https://xxx.fusionsolar.huawei.com/thirdData/login** (Enter the actual URL of the NMS.)
3. Request body: API account name and password (Use **systemCode** instead of **password** as the password parameter.)
4. HTTP status code: The response status code 200 indicates that the API is successfully called.
5. Response body: If **failCode** is 0, the API login is successful.
6. Header: After successful login, find the **xsrftoken** field in the response header and its value is the credential information required for accessing northbound APIs.

NOTICE

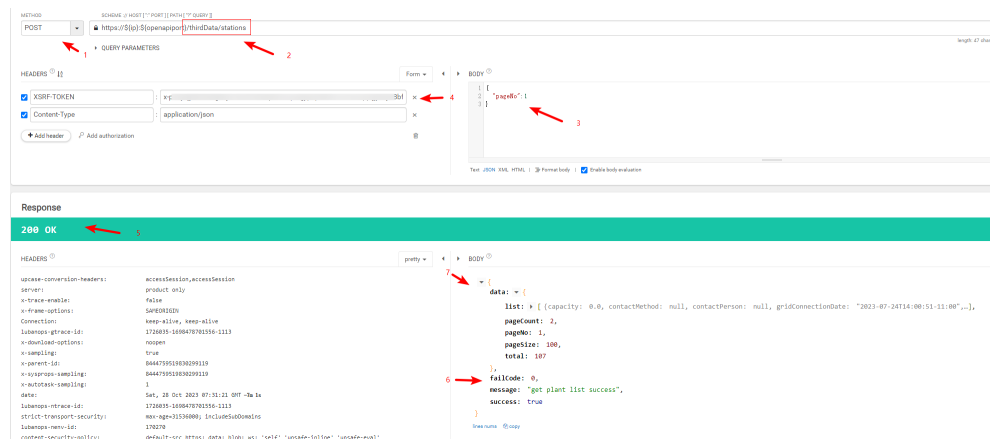
1. The validity period of a token is 30 minutes. If you continuously call a token within 30 minutes, the validity period will be automatically extended. Therefore, you must keep the token secure carefully to prevent disclosure.
 2. An account can have only one online session. Repeated login will invalidate the generated token. If a token is invalid, check to eliminate the repeated login issue.
-



7.2 Querying the Plant List

Before querying the plant information, you need to query the plants on which the current API account has the permission as follows:

1. Method: POST
2. URL: **`https://xxx.fusionsolar.huawei.com/thirdData/stations`** (Enter the actual URL of the NMS.)
3. Request body: Enter the page No. of the plant list API. The following example queries page No. 1. By default, a maximum of 100 records can be returned for each page.
4. XSRF-TOKEN: Enter the token information obtained through the **`/thirdData/login`** API.
5. HTTP status code: The response status code 200 indicates that the API is successfully called.
6. Response body: If **failCode** is 0, the query is successful.
7. Response body: The **data** field indicates the information about the plants that you have permission to query.



NOTE

Expand the details in the **data** field to obtain the basic information about a plant, including **plantCode**, the unique identifier for data query. You can save the information for future use.

```
{
  "data": {
    "list": [
      {
        "capacity": 0.0, "contactMethod": null, "contactPerson": null, "gridConnectionDate": "2023-07-24T14:00:51-11:00", ...
      }
    ]
  }
}
```

7.3 Querying the Real-Time Plant Data

You can query the real-time and historical data and reports of plants. The method to obtain real-time plant data is as follows:

1. Method: POST

2. URL: **`https://xxx.fusionsolar.huawei.com/thirdData/getStationRealKpi`** (Enter the actual URL of the NMS.)

3. Request body: Enter the value in **plantCode** obtained from the plant list API. If multiple values are used, separate them using commas (,). A maximum of 100 values are supported.

4. XSRF-TOKEN: Enter the token information obtained through the **`/thirdData/login`** API.

5. HTTP status code: The response status code 200 indicates that the API is successfully called.

6. Response body: If **failCode** is 0, the query is successful.

7. Response body: The queried real-time plant data is shown in the **data** field.